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Underground Mining and Bulk Materials Handling
(MIN E – 630)
Assignment #2

1 Validate the numbers of “Remaining Movable Resources i” in Sequences #2 and #3

Based on the preset information. At the beginning, we have 8200000 tons resources and two stopings. The first sequence includes Year-3, Year-4, and Year-5. So, the resource that will be mined can be calculated as follows:

$$\text{amount of sequence 1} = \text{daily production} \times \text{\#of year} \times 320 = 768000$$

Therefore, for sequence 2, the remaining movable resource is equal to $8200000 - 768000 = 7432000$.

Similarly, the resource that will be mined in sequence 2 can be calculated as follows:

$$\text{amount of sequence 1} = \text{daily production} \times \text{\#of year} \times 320 = 1600 \times 8 \times 320 = 409600$$

Thus, for sequence 3, the remaining movable resource is equal to $7432000 - 4096000 = 3336000$. The numbers of “Remaining Movable Resources” in Sequence #2 and Sequence #3 are correct.

2 In the software chat, there is a relationship between “Resource size, tons” and “Production, tpd.” Please use hand calculations to verify the relationship. Is this relationship accurate?

Enter the total amount of ore excavated and transported daily. Guideline production rates calculated using Taylor's Rule (an empirical formula sometimes used to relate min life to resource size) are:

Resource size, tons.....	Production, tpd
100,000.....	Ships 80 y
1,000,000.....	450
5,000,000.....	1,500
10,000,000.....	2,500
50,000,000.....	8,400
100,000,000.....	14,000
500,000,000.....	46,000

Figure 1: The relationship between resource and production

According to the tip information, the guideline production rates are calculated using Taylor's Rule (see Equation 2).

Substitute the resources,

$$\text{Life of Mine} = 0.2 \times 8200000^{0.25} \pm 1.2 = 10.7 \text{ year}$$

This leads to an estimate of the daily production:

$$\text{Daily production} = \frac{\text{Resource}}{\text{Life of Mine} \times 320} = \frac{8200000}{10.7 \times 320} = 2395$$

We use the interpolation method to get the production from this table. The formula for linear interpolation is:

$$y = y_1 + \frac{x - x_1}{x_2 - x_1} \times (y_2 - y_1) \quad (1)$$

Substitute the values into the Equation 1:

$$y = 1,500 + \frac{3,200,000}{5,000,000} \times (1,000) = 2140$$

As we can see, there is a bias between 2395 and 2140. I think this is because we set a different number of workdays. If we take the number of workdays as 358 days, we can get these two numbers very close.

3 Verify Taylor's Rule

$$\text{Life of Mine} = 0.2 \times (\text{expected ore tonnes})^{0.25} \pm 1.2 \text{ years} \quad (2)$$

For sequence #1, the life of mine under Taylor's rule,

$$\text{Life of Mine} = 0.2 \times 768000^{0.25} \pm 1.2 = 5.9 \text{ years}$$

For sequence #2, the life of mine under Taylor's rule,

$$\text{Life of Mine} = 0.2 \times 4096000^{0.25} \pm 1.2 = 9.0 \text{ years}$$

For sequence #3, the life of mine under Taylor's rule,

$$\text{Life of Mine} = 0.2 \times 3336000^{0.25} \pm 1.2 = 8.55 \text{ years}$$

In our project, for sequence #1, #2, and #3, we planned 3 years, 8 years, and 2 years for each sequence, respectively. These numbers don't match Taylor's Rule.

4 As a design engineer, how do you decide the "Life of Mine"? Please have a discussion.

As a design engineer, I should have a comprehensive understanding of each project before we decide the "Life of Mine".

First, we should refer to the empirical formula (e.g., Taylor's Rule) or the history database to obtain a rough life of mine.

And then, we should collect as much detailed data as possible, such as the detailed geological conditions and the equipment and technical constraints. Using these conditions as some justification, variables adjust the rough life of mine obtained before.

Finally, we should also pay attention to the mining market based on economic considerations.

In conclusion, we should keep a consecutive dynamic action for optimizing the mining schedule to get the maximum profit as much as possible.

5 Add Adit #2 using a similar dimension (compared with Adit #1) and 500 ft rock bolts at the “Portal Elevation” of 1300 ft.

The parameters of Adit #2 is shown in the Figure 2.

The screenshot shows a software interface for configuring 'Adit 2'. At the top, there are tabs for 'ADIT 1', 'ADIT 2' (which is selected and highlighted in green), 'ADIT 3', and 'ADIT 4'. Below the tabs, the title 'Adit 2' is followed by a green pencil icon and the text 'Daily Advance: 10.31 Feet/Day'. The main configuration area is titled 'Adits' and contains a grid of input fields for various parameters. Each field has a question mark icon for help. The parameters and their values are as follows:

Parameter	Value	Unit
Location	Footwall	
Footwall	Footwall	
Face Width	14.17	Feet
Face Height	11.96	Feet
Length	6,600	Feet
Gradient	1.00	Percent
Year Begun	4	
Year Complete	5	
Sequence Efficiency	95.00	Percent
Rock Bolts	500	Feet
Timber	0	Feet
Lagging	0	Feet
Shotcrete	6,600	Feet
Concrete	0	Feet
Steel Liner Plate	0	Feet
Portal Elevation	1,300	Feet

Figure 2: The parameters of Adit#2

6 Add one shaft using the following settings:

The paramertes of the shaft is shown in the Figure 3.

SHAFT 1
SHAFT 3
SHAFT 4

Shaft 1
Daily Advance: 0 Feet/Day

Ore

Plan Section
Circular
Circular
Location
Footwall
Footwall
Hoist
Production
Production

Number of Compartments
1
Face Area
1,500.00
Square Feet
Depth
2,920
Feet

Year Begun
12
Year Complete
13
Sequence Efficiency
72.00
Percent

Rock Bolts
0
Feet
Timber
0
Feet
Lagging
0
Feet

Steel Liner PLate
0
Feet
Concrete
2,920
Feet
Collar Elevation
2,500
Feet

Figure 3: The parameters of Shaft

7 Report the “Capital Costs” to build Adit#2 and the shaft

The capital costs to build Adit #2 and the shaft are shown in the Table 1.

Table 1: Capital Costs

name	Year							
	Year 1	Year 2	3	Year 4	Year 5	...	Year 12	Year 13
Shaft #1	\$0	\$0	\$0	\$0	\$0	...	\$21,775,741.01	\$21,775,741.01
Adit #1	\$4,915,392.93	\$4,915,392.93	\$0	\$0	\$0	...	\$0	\$0
Adit #2	\$0	\$0	\$0	\$4,805,504.19	\$4,805,504.19	...	\$0	\$0
Drifts	\$0	\$600,100.10	\$0	\$0	\$0	...	\$0	\$0

8 What are the Total Capital Cost and Total Operating Cost for the whole project?

The executive summary of the total project cost is shown in the following.

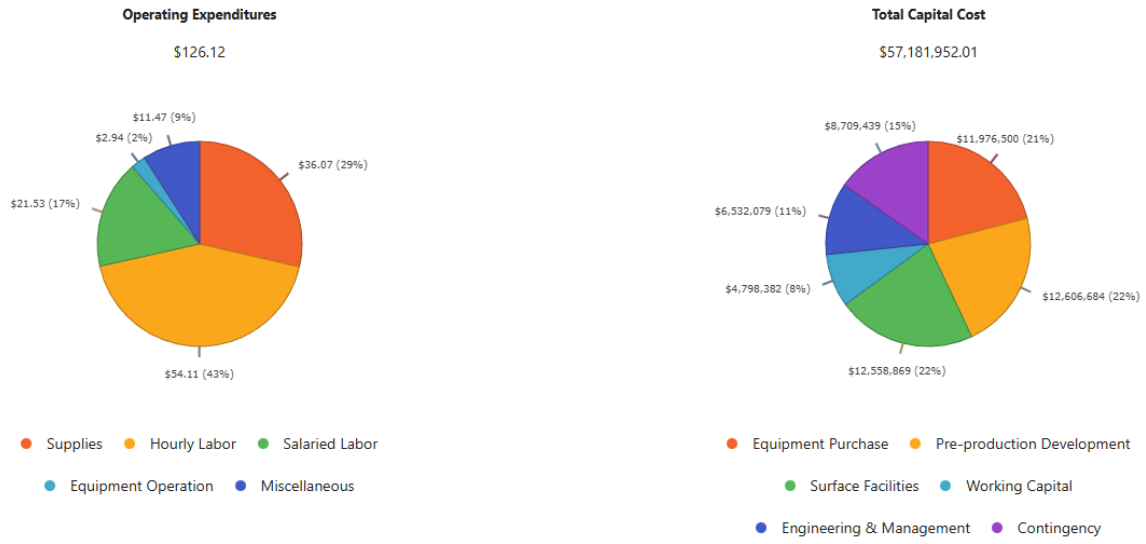


Figure 4: The sequence 1 of Executive summary

From Figure 4, we plan to mine 76800 tonnes, so the total operating cost of sequence 1 is:

$$768000 \times 126.12 = 96860160$$

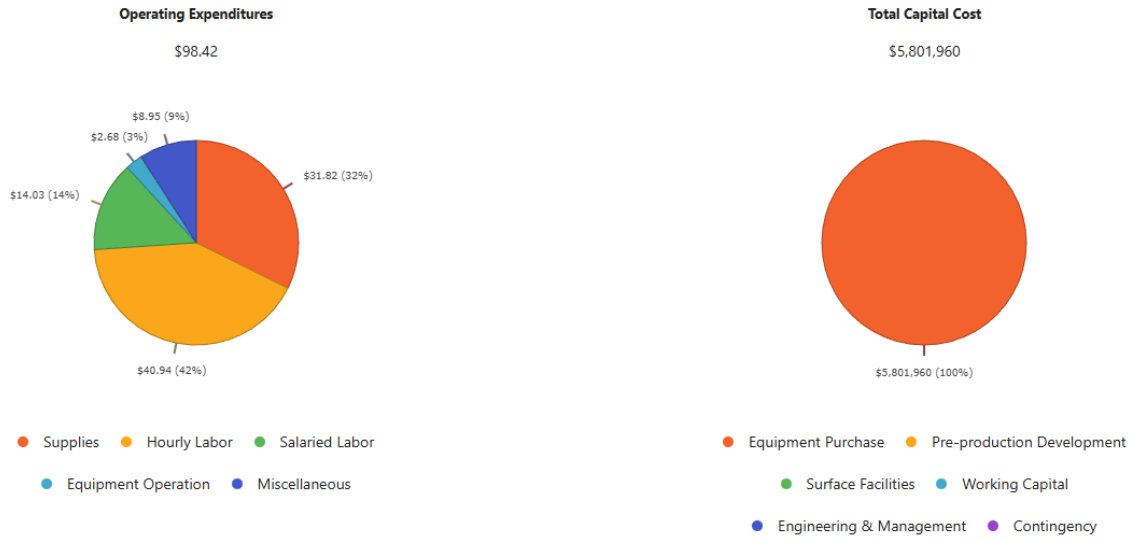


Figure 5: The sequence 2 of Executive summary

From Figure 5, we plan to mine 76800 tonnes, so the total operating cost of sequence 2 is:

$$4096000 \times 98.42 = 403128320$$



Figure 6: The sequence 3 of Executive summary

From Figure 6, we plan to mine 76800 tonnes, so the total operating cost of sequence 3 is:

$$3336000 \times 98.04 = 327061440$$

Thus, the total operating costs are $327061440 + 403128320 + 96860160 = 827049920$.

The total capital costs are $131600 + 5801960 + 57,181,952.01 = 63115512.01$

Based on the total project cost summary, the capital costs are listed as annual expenditures across the 22-year project timeline. Summing the values as follows: which includes equipment purchase, pre-production development, surface facilities, working capital, engineering, management, and contingency)

The “Total Operating Costs” sums the individual components: supplies, labor, equipment operation, and miscellaneous for each year of production (Years 3 to 22).